CHILD, MATERNAL & ADOLESCENT HEALTH OUTREACH

Community Health & Outreach Management System (CHOMS)

Replacing paper registers with a secure, offline-first cloud platform.



MatCH (Maternal, Adolescent and Child Health Institute) | Work Integrated Learning 3 (XADAD7112w)

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Every child counted. Every family reached.

The problem today

Paper registers
duplicate clients, no history

Lost referrals
no outcome tracking

Delayed visibility
coordinators learn weeks later

Manual funder reports
compiled by hand every cycle

Our solution

One secure cloud backend
Firebase Auth + Firestore

3 front ends
Website, installable PWA, Android app

Real-time dashboards
live work lists and charts

Offline-first sync
capture now, sync when signal returns

MatCH community health workers visit mothers, children and adolescents across Durban every day.

WHAT WE DELIVERED

One backend, three matching front ends

Public website

Landing page with Create account and Sign in. Same child-welfare design as the apps.

Web / desktop app (PWA)

Installable, works offline. Full system for CHWs, coordinators and admins.

Native Android app

Java, MVVM + Repository, Material 3. Firebase Auth and Firestore.

Firebase project: match-choms

Authentication + Cloud Firestore (africa-south1, Johannesburg) + server-side Security Rules

HOW IT IS BUILT

System Architecture (N-Tier)



Presentation

HTML/CSS/JS (PWA) + Native Android (Java, MVVM, Material 3)



Business Logic

Pure JavaScript (utils.js: validation, SA ID, permissions) + Android ViewModels



Data Access

Repository Pattern (repo.js): role-scoped reads, cache, outbox, audited batch writes



Cloud

Firebase Auth, Cloud Firestore (africa-south1) and server-side Security Rules

Why layers?

- Each layer has one job
- UI never talks to Firestore directly
- Rules enforce the same permissions on the server
- Logic is plain JavaScript, so it is unit-tested with Jest

ROLE-BASED ACCESS CONTROL

Secure by Design

Admin

Full system access, user management, approve reports

Coordinator

View all data, generate and save reports

CHW (Community Health Worker)

See and edit only their assigned clients and visits

Pending

New sign-ups must be approved by an Admin; no data access until approved

Account lifecycle

Sign up → **Pending** → **Admin approves + sets role** → **Active** → **Deactivate / delete (audited)**

THE IMMUTABLE AUDIT TRAIL

Zero Unaudited Writes

Paired writes

Every create, update and delete is committed in one atomic batch with an auditLogs entry.

Server-enforced

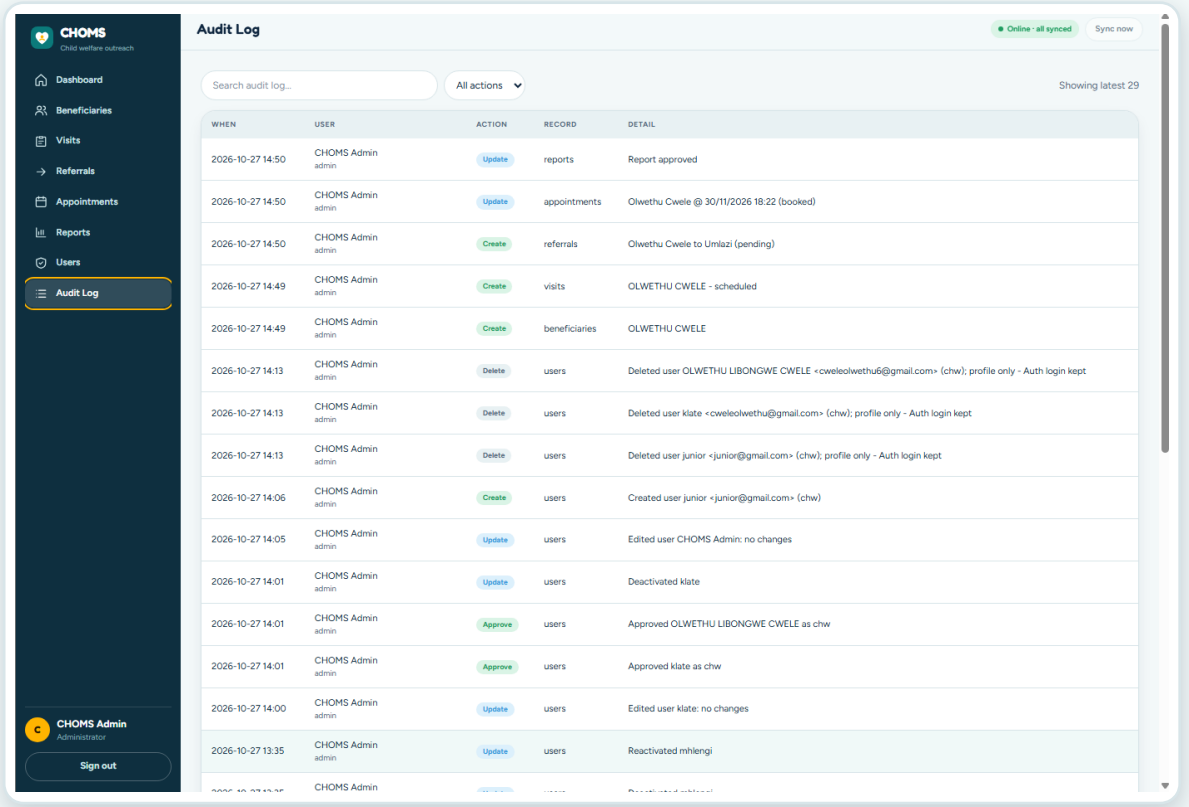
Firestore rules reject the write if the audit entry is missing.

Immutable

Nobody, including Admins, can edit or delete an audit log entry.

Accountable

Who did what and when, for sensitive child health data (POPIA).



Audit log screen: searchable by user and action

OFFLINE-FIRST SYNCHRONISATION

Working Without Signal



No internet

CHWs work in areas with no signal



Local outbox

Data saved to localStorage; service worker caches the app shell



Reconnect

Outbox flushes automatically (about every 60 s, or Sync Now)



Firebase

Synced with its audit entry; failures are flagged, never lost

Pending changes are always visible: a "Sync Now" chip shows how many records are waiting to upload.

KEY FEATURES AND UI/UX

Built for the Field and the Office

SA ID validation

Auto-fills date of birth and gender, blocks duplicates

Dashboards

Stat cards, work lists, completed visits per CHW chart

Reports

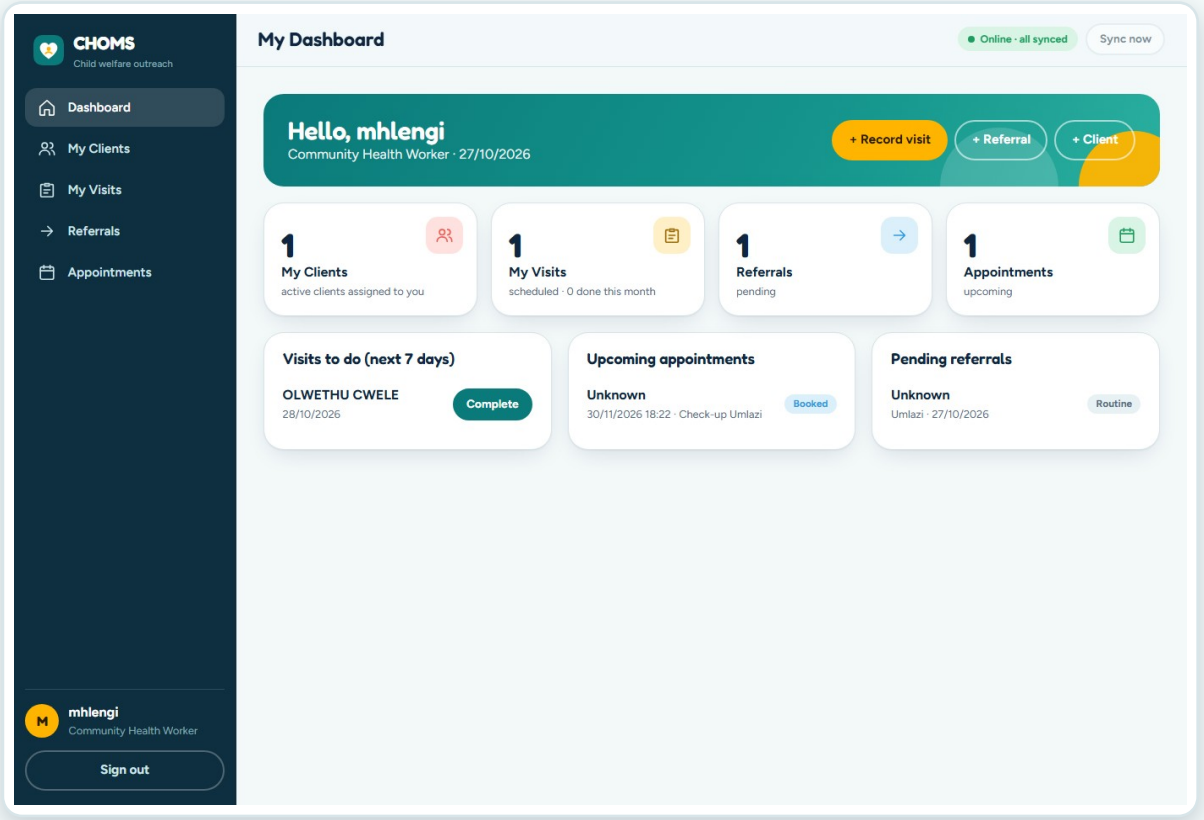
Any date range, export CSV, print / PDF

User management

Approve, edit, deactivate and delete accounts

Friendly errors

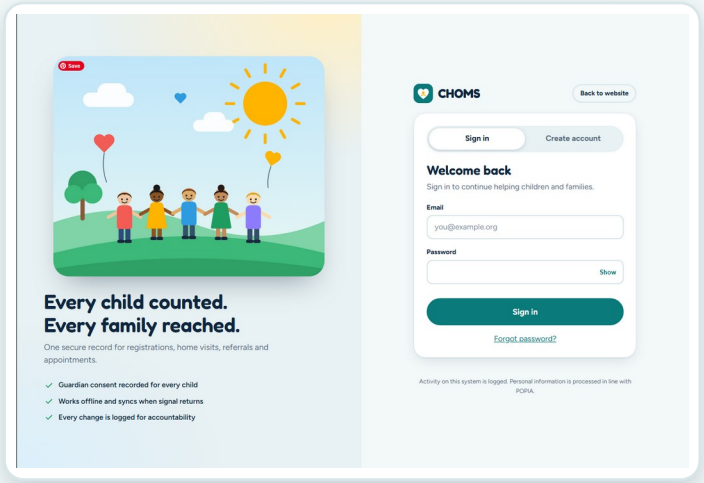
Plain-language messages, no technical jargon



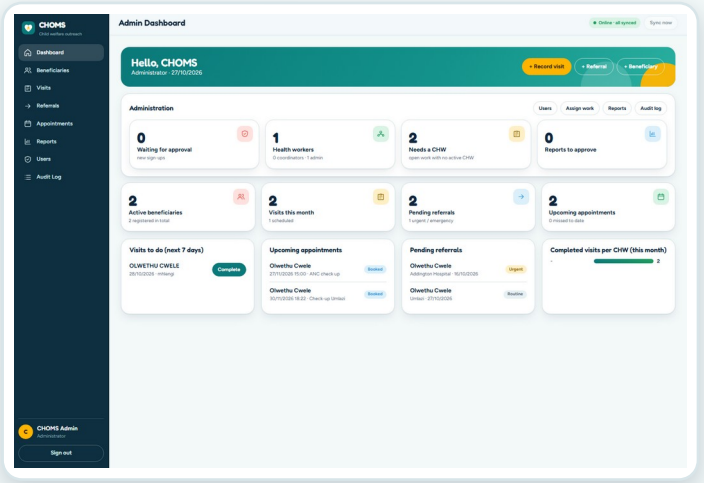
CHW dashboard: my clients, visits and referrals at a glance

SCREENSHOTS

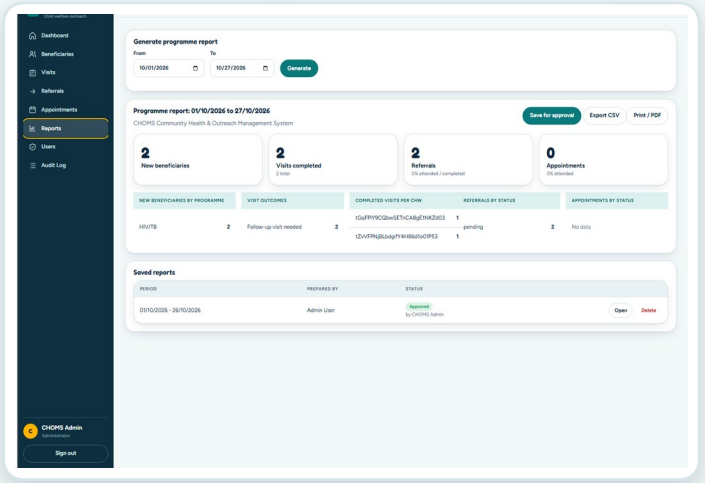
The system in action



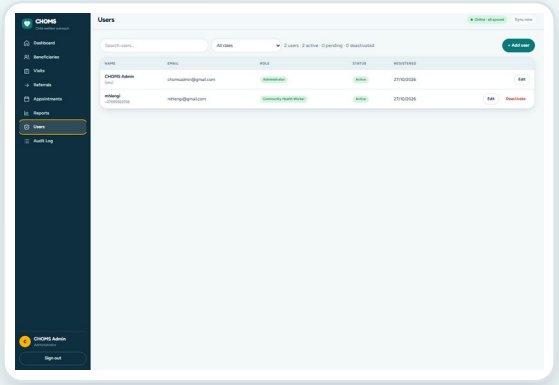
Sign in



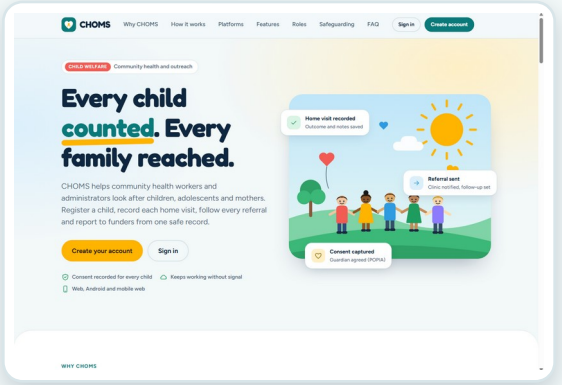
Admin dashboard



Reports



User management



Public website

DEVOPS AND TESTING

Verified, Not Assumed

50

Jest unit tests

Business logic: validation, SA ID, permissions

21

JUnit tests

Android app validators and logic

76

Security rule tests

Simulating real attacks on Firestore rules

GitHub Actions CI pipelines automatically test the code on every push

CHALLENGES AND SOLUTIONS

Overcoming Real-World Hurdles



User deletion without a paid Cloud Function

Built a secure admin-only client fallback that removes the profile, revokes access and logs the deletion to the audit trail.



UI white space bug

Fixed CSS container constraints so wide screens get a full-width enterprise layout.



Sign-in loop

Fixed caching issues by always fetching fresh profiles from the server, so pending users cannot loop.

Transforming Health Outreach

CHOMS eliminates paper trails, prevents lost referrals, and ensures 100% data accountability.

Future work

SMS gateway

Appointment reminders

Native iOS app

Same rules and audit contract

Offline-first Android UI

Local queue with pending indicator

Provincial integration

Export to health information systems

THANK YOU

Our Team & Live System



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Firebase project: match-choms

Website: http://127.0.0.1:8123/index.html

Web app: http://127.0.0.1:8123/app/index.html

GitHub: github.com/Olwethu7/choms-group-work

Firebase: console.firebase.google.com/project/match-choms